DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

# WO-100-04 — Remake flanged joint 2"-P300-LS-3007/A2 on 2"-P300-LS-3007

| | | | |
|---|---|---|---|
| **Area** | A100 Tank farm | **Task** | GJ Flanged joint remake |
| **Priority** | 1 (safety) | **Issued by** | Facility owner (demonstration) |
| **Primary asset** | T-101 — T-101 tank 12' dia × 22' high — E 25'-0"  N 26'-6"  EL 100'-6" | | |
| **Equipment** | T-101 | **Lines** | 2"-P300-LS-3007 |
| **Isometrics** | ISO-LS-3007 (owner's drawings, attached) | | |
| **Permits** | General work permit | | |

## Reason

Steam leak at the tank heating-coil inlet flange.

## Scope of work

1. Isolate per the isolation list, drain and vent 2"-P300-LS-3007.
2. Break the joint at E 22'-0"  N -12"  EL 123'-9-1/4" between FLANGE WELD-NECK 2" P300 RAISED FACE and VALVE, GATE, WEDGE, OS&Y 2" P300.
3. Inspect both flange faces (SW-NDE-5).
4. Fit a new gasket and new studs; tighten to the stated preload in three passes, cross pattern (SW-WO-5).
5. Remove blinds; leak test on start-up (SW-TEST-2).

Affected items: 2"-P300-LS-3007/A2, 2"-P300-LS-3007/A1, 2"-P300-LS-3007/A3

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-5, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-7

## Owner's isometric drawings (SW-IMP-6)

| Drawing | Line drawn | Work line | Sheets | Match to the plant model |
|---|---|---|---|---|
| ISO-LS-3007 | 2"-P300-LS-3007 | 2"-P300-LS-3007 | 1 | agrees with the model |

ISO-LS-3007 (PDF): ../isometrics-imported/ISO-LS-3007.pdf — the sheets that hold the work follow the ITP.

## Assets in the plant model (SW-WO-7)

| Tag | Kind | Description | Line | Location |
|---|---|---|---|---|
| T-101 | equipment | T-101 tank 12' dia × 22' high | | E 25'-0"  N 26'-6"  EL 100'-6" |
| 2"-P300-LS-3007/A2 | item | GASKET 2" P300 COMPRESSED SHEET | 2"-P300-LS-3007 | E 22'-0"  N -12"  EL 123'-9-1/4" |
| 2"-P300-LS-3007/A1 | item | FLANGE WELD-NECK 2" P300 RAISED FACE | 2"-P300-LS-3007 | E 22'-0"  N -12"  EL 123'-7-1/8" |
| 2"-P300-LS-3007/A3 | item | VALVE, GATE, WEDGE, OS&Y 2" P300 | 2"-P300-LS-3007 | E 22'-0"  N -12"  EL 124'-2-11/16" |
| 6"-P300-LS-2003/T:2"-P300-LS-3007 | item | TEE REDUCING 6" × 2" P300 | 6"-P300-LS-2003 | E 22'-0"  N -12"  EL 123'-0" |
| 2"-P300-LS-3007/B2 | item | VALVE, GATE, WEDGE, OS&Y 2" P300 | 2"-P300-LS-3007 | E 18'-5/8"  N 26'-6"  EL 102'-6" |
| 2"-P300-LS-3007/B5 | item | FLANGE WELD-NECK 2" P300 RAISED FACE | 2"-P300-LS-3007 | E 17'-5-1/16"  N 26'-6"  EL 102'-6" |

## Isolation list (SW-WO-3)

| # | Kind | Tag | Line | Action | Location | Fitted ✓ | Removed ✓ |
|---|---|---|---|---|---|---|---|
| 1 | equipment | 6"-P300-LS-2003/T:2"-P300-LS-3007 | 6"-P300-LS-2003 | No valve on this side within the line: isolate or depressure the parent line 6"-P300-LS-2003 at its tee under the shutdown isolation | | | |
| 2 | valve | 2"-P300-LS-3007/B2 | 2"-P300-LS-3007 | Close and lock VALVE, GATE, WEDGE, OS&Y 2" P300 (2"-P300-LS-3007/B2) | E 18'-5/8"  N 26'-6"  EL 102'-6" | | |
| 3 | blind | 2"-P300-LS-3007/B5 | 2"-P300-LS-3007 | Fit a line blind between 2"-P300-LS-3007/B2 and FLANGE WELD-NECK 2" P300 RAISED FACE (2"-P300-LS-3007/B5), work side | E 17'-7-1/8"  N 26'-6"  EL 102'-6" | | |

## Parts

| Description | Qty | Replaces |
|---|---|---|
| GASKET 2" P300 COMPRESSED SHEET | 1 | 2"-P300-LS-3007/A2 |
| STUD SET 4 × 0.625" × 4-1/4" WITH 2 NUTS, FOR 2" P300 | 1 | 2"-P300-LS-3007/A4 |

## Inspections

| Id | Inspection | Criteria | Basis |
|---|---|---|---|
| WO-100-04-I1 | Flange faces before the new gasket goes in | No radial scratch across the sealing face (SW-NDE-5) | SW-NDE-5 |

## Bolt-up (SW-WO-5)

| Joint | Studs | Preload per stud (lbf) | Passes | Done ✓ |
|---|---|---|---|---|
| 2"-P300-LS-3007/A2 | 4 × 0.625" studs (2"-P300-LS-3007/A4) | 13,806 | 3 | |

## Test — Service leak test

| | |
|---|---|
| **Description** | The remade joint checked at operating pressure on start-up |
| **Basis** | SW-TEST-2 |

## Inspection and test plan (SW-ITP-1..3)

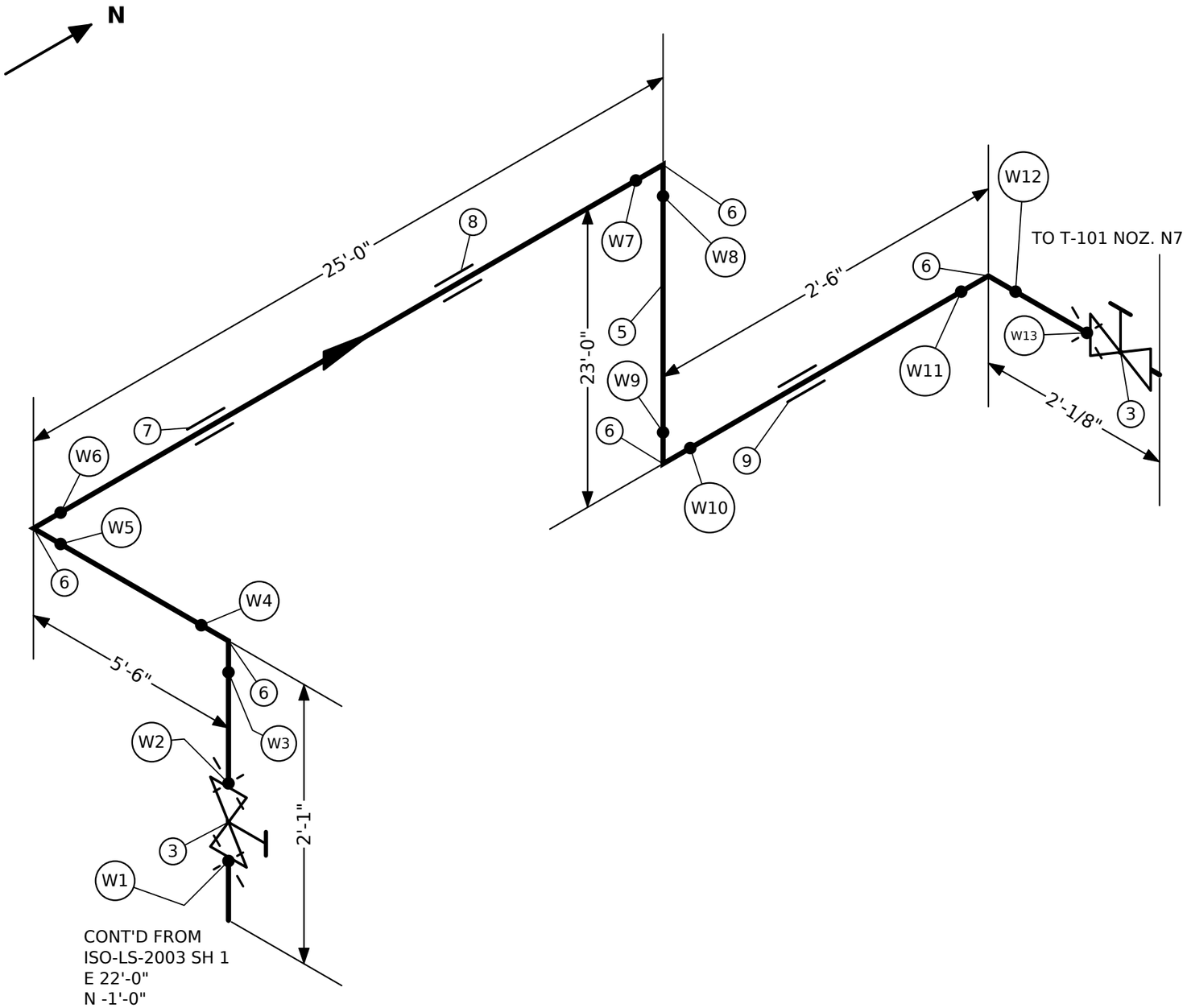
H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

| # | Activity | Reference | Acceptance | Record | Contr. | Owner | 3rd party | Status | Signed (contr. / owner / 3rd) | Date | Remark |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained | SW-WO-3, isolation list | Every isolation in place and tagged; zero pressure at the vent | Signed isolation certificate | H | H | — | | | | |
| 2 | Joint broken; both flange faces inspected | SW-NDE-5 | No radial scratch across the sealing face | Flange face inspection record | W | W | — | | | | |
| 3 | New gasket fitted; studs tightened to preload in three passes | SW-WO-5 | Preload per stud as stated | Bolt-up record | W | W | — | | | | |
| 4 | Service leak test on start-up | SW-TEST-2 | No leak | Leak test record | W | W | — | | | | |
| 5 | Blinds removed, isolations returned to normal line-up per the isolation list | SW-WO-3 | Every blind removed and accounted for | Isolation certificate closed | H | H | — | | | | |
| 6 | Records complete; work order handed back to operations | SW-ITP-3 | All ITP activities accepted or not applicable | Signed ITP and completion record | H | H | R | | | | |

## Where the work is on the owner's drawings (SW-IMP-8)

Each affected item, replaced weld and isolation point, with the drawing and sheet that hold it and the weld label printed there. The sheets follow, copied unchanged.

| Tag | Role | Drawing | Sheet | Weld label |
|---|---|---|---|---|
| 2"-P300-LS-3007/A2 | affected | ISO-LS-3007 | 1 | |
| 2"-P300-LS-3007/A1 | affected | ISO-LS-3007 | 1 | |
| 2"-P300-LS-3007/A3 | affected | ISO-LS-3007 | 1 | |
| 6"-P300-LS-2003/T:2"-P300-LS-3007 | isolation: equipment | ISO-LS-2003 | 1 | |
| 2"-P300-LS-3007/B2 | isolation: valve | ISO-LS-3007 | 1 | |
| 2"-P300-LS-3007/B5 | isolation: blind | ISO-LS-3007 | 1 | |




## BILL OF MATERIALS

| PT | QTY | NPS | DESCRIPTION |
|---|---|---|---|
| 1 | 3 | 2" | FLANGE WELD-NECK 2" P300 RAISED FACE |
| 2 | 4 | 2" | GASKET 2" P300 COMPRESSED SHEET |
| 3 | 2 | 2" | VALVE, GATE, WEDGE, OS&Y 2" P300 |
| 4 | 4 | 2" | STUD SET 4 × 0.625" × 4-1/4" WITH 2 NUTS, FOR 2" P300 |
| 5 | 54'-5 1/4" | 2" | PIPE 2" P300 BEVELLED ENDS |
| 6 | 5 | 2" | ELBOW 90° R = 5/3 OD 2" P300 |
| 7 | 1 | 2" | HANGER FOR 2" LINE (HANGER-2x87.0) |
| 8 | 1 | 2" | HANGER FOR 2" LINE (HANGER-2x77.0) |
| 9 | 1 | 2" | TRUNNION FOR 2" LINE (TRUNNION-2x29.0) |

| REV | DATE | DESCRIPTION | BY |
|---|---|---|---|
| 0 | 2026-11-01 | ISSUED FOR CONSTRUCTION | - |
| 1 | 2026-11-02 | REVISED TO MATCH THE MODEL (FL-1) | - |

**NOTES:**

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. HYDROTEST BEFORE PAINTING
4. CHECK ORIENTATION BEFORE WELDING

| TEST PRESS. | INSULATION | PAINT |
|---|---|---|
| - | YES | - |
| PWHT | NDE | FLUID |
| - | - | LS |

COMPANY

**SMARTWELD**

TITLE

**LOW-PRESSURE STEAM**

| LINE NO. | PIPING CLASS | DESIGN PRESS. | DESIGN TEMP. |
|---|---|---|---|
| 2"-P300-LS-3007 | S300 | 300 PSIG | - |

| DRAWING NO. | SHEET | REV |
|---|---|---|
| ISO-LS-3007 | 1 OF 1 | 1 |

| DATE | DRAWN | CHECKED | SCALE |
|---|---|---|---|
| 2026-11-01 | - | - | NTS |

# Notices

**DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.**

This document embeds a subset of the DejaVu Sans font (SW-LIC-3), renamed under its licence. DejaVu Sans (assets/fonts/DejaVuSans.ttf), from the DejaVu fonts 2.37 as packaged in the build environment.

Copyright (c) 2003 by Bitstream, Inc. All Rights Reserved. Bitstream Vera is a trademark of Bitstream, Inc.
DejaVu changes are in public domain.

Permission is hereby granted, free of charge, to any person obtaining a copy
of the fonts accompanying this license ("Fonts") and associated
documentation files (the "Font Software"), to reproduce and distribute the
Font Software, including without limitation the rights to use, copy, merge,
publish, distribute, and/or sell copies of the Font Software, and to permit
persons to whom the Font Software is furnished to do so, subject to the
following conditions:

The above copyright and trademark notices and this permission notice shall
be included in all copies of one or more of the Font Software typefaces.

The Font Software may be modified, altered, or added to, and in particular
the designs of glyphs or characters in the Fonts may be modified and
additional glyphs or characters may be added to the Fonts, only if the fonts
are renamed to names not containing either the words "Bitstream" or the word
"Vera".

This License becomes null and void to the extent applicable to Fonts or Font
Software that has been modified and is distributed under the "Bitstream
Vera" names.

The Font Software may be sold as part of a larger software package but no
copy of one or more of the Font Software typefaces may be sold by itself.

THE FONT SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS
OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTIES OF MERCHANTABILITY,
FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT OF COPYRIGHT, PATENT,
TRADEMARK, OR OTHER RIGHT. IN NO EVENT SHALL BITSTREAM OR THE GNOME
FOUNDATION BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, INCLUDING
ANY GENERAL, SPECIAL, INDIRECT, INCIDENTAL, OR CONSEQUENTIAL DAMAGES,
WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF
THE USE OR INABILITY TO USE THE FONT SOFTWARE OR FROM OTHER DEALINGS IN THE
FONT SOFTWARE.

Except as contained in this notice, the names of Gnome, the Gnome
Foundation, and Bitstream Inc., shall not be used in advertising or
otherwise to promote the sale, use or other dealings in this Font Software
without prior written authorization from the Gnome Foundation or Bitstream
Inc., respectively. For further information, contact: fonts at gnome dot
org.